

PROJECT PROPOSAL

A. GENERAL INFORMATION

A1. Project Title

"Digital Twin Reconstruction of Base Transceiver Stations with 3D Gaussian Splatting: Studying Vanilla 3DGS and Building an Improved 3DGS Variant Better Suited to BTS Towers"

A2. Duration

3 months .

A3. Principal Investigator

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Student ID: ITCSIU24047

University: International University – Vietnam National University Ho Chi Minh City

A4. Project Members (including the Principal Investigator)

No.	Full name	Student ID	University / School
1	Lê Anh Kiệt	ITCSIU24047	International University – VNU-HCM

B. RESEARCH DESCRIPTION

B1. Introduction

This is a course project. Its goal is to study how the original (vanilla) 3D Gaussian Splatting (3DGS) pipeline works, use it to reconstruct a real Base Transceiver Station (BTS) telecom tower as a digital twin from drone photographs, and look for a rendering method that works better than vanilla 3DGS for this kind of scene. A BTS scene is mostly flat surfaces (roofs, walls, sky), but the tower mast, antennas, and cables are thin structures that are harder to render well.

Figure 1 shows a real result from our pipeline: a novel view of a BTS tower rendered by the trained 3DGS model, from drone photographs of a rooftop site in Ho Chi Minh City.



Figure 1: Output example: a BTS tower rendered by the trained 3DGS model from a new camera view.

B2. Objectives and Research Content

- Study the original 3DGS pipeline and its training/density-control bottlenecks as the baseline.
- Survey multiple published techniques for faster training and better densification, and verify each one against its own source code. We will focus on techniques that improve rendering of thin structures (the tower mast, antennas, cables), such as anti-aliasing, opacity/depth regularization, and density control for sparse regions.
- Select and combine the most effective techniques into one optimized pipeline for BTS digital-twin reconstruction.

- Benchmark the combined pipeline against vanilla 3DGS using PSNR, SSIM, LPIPS, Gaussian count, training time, and memory use.

B3. Expected Results

- A short, verified report and slides comparing multiple 3DGS optimization techniques.
- An optimized 3DGS pipeline for BTS digital twins, combining the best-performing techniques.
- A real benchmark table comparing the optimized pipeline with vanilla 3DGS.